

FORENSIC SCIENCE LABORATORY REPORT

Case: FIR/2024/DL/PSRKP/0452

Date: 13-08-2026

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Fsl Report Number: FSL/DL/2024/PHY/8934

Date Of Receipt: 2024-10-17

Date Of Examination: 2024-10-22 to 2024-11-15

Date Of Report: 2024-11-18

Laboratory: Forensic Science Laboratory, Rohini, New Delhi - 110085

EXAMINING OFFICER

Name: Dr. Priya Nambiar

Designation: Senior Scientific Officer (Physics)

Qualification: M.Sc. Physics, Ph.D. Forensic Science

CASE REFERENCE

Fir Number: FIR/2024/DL/PSRKP/0452

Police Station: Rajinder Nagar

Forwarding Authority: Inspector Suresh Chandra

EXHIBITS RECEIVED

Item 1:

Exhibit No: EX-01

Evidence Id: EVD-0452-001

Description: Bajaj Pulsar 220F Motorcycle DL2SBP1234

Parcel Condition: Intact, sealed with seal 'SC'

Item 2:

Exhibit No: EX-02

Evidence Id: EVD-0452-002

Description: Honda Activa 6G Scooter DL5SCT5678

Parcel Condition: Intact, sealed with seal 'SC'

Item 3:

Exhibit No: EX-03

Evidence Id: EVD-0452-006

Description: Helmet - Accused (Steelbird, Black)

Parcel Condition: Intact, sealed

Item 4:

Exhibit No: EX-04

Evidence Id: EVD-0452-007

Description: Helmet - Victim (Vega, White)

Parcel Condition: Intact, sealed

Item 5:

Exhibit No: EX-05

Evidence Id: EVD-0452-008

Description: Broken vehicle parts from site

Parcel Condition: Intact, sealed

Item 6:

Exhibit No: EX-06

Evidence Id: EVD-0452-010

Description: Blood sample - Accused (BLD-0452-A)

Parcel Condition: Intact, refrigerated transport

Item 7:

Exhibit No: EX-07

Evidence Id: EVD-0452-011

Description: Blood sample - Victim (BLD-0452-B)

Parcel Condition: Intact, refrigerated transport

Item 8:

Exhibit No: EX-08

Evidence Id: EVD-0452-012

Description: Digital photographs (SD Card, 32GB)

Parcel Condition: Intact, sealed

Item 9:**Exhibit No:** EX-09**Evidence Id:** EVD-0452-013**Description:** CCTV DVR Hard Disk (1TB)**Parcel Condition:** Intact, sealed**EXAMINATIONS CONDUCTED****VEHICLE EXAMINATION****Method:** Physical inspection, measurement, photographic documentation**FINDINGS PULSAR****Speedometer:** Needle stuck at 55 kmph (indicates approximate speed at impact)**Headlight:** Smashed, filament examination shows 'hot shock' deformation confirming light was ON at time of impact**Front Wheel:** Rim bent, tyre burst on right side**Brake System:** Front disc brake functional, rear drum brake functional. Brake lever in applied position consistent with emergency braking**Impact Force Calculated:** Approximately 18-22 kN based on deformation pattern**Paint Evidence:** White paint transfer on right front fork matching Activa body color**FINDINGS ACTIVA****Right Side Impact:** Deformation depth 25cm on right body panel indicating lateral impact force**Paint Evidence:** Red paint transfer on right side panel matching Pulsar color**Brake Lever:** Right brake lever bent due to impact, position suggests braking attempt**Tyres:** Both tyres in good condition, adequate tread depth (3.5mm front, 4.0mm rear)**Conclusion:** Primary impact was between front-right portion of Pulsar and right-center portion of Activa at an angle of 70-80 degrees. Pulsar was traveling at minimum 50-55 kmph at moment of impact. Both vehicle brake systems were functional prior to accident. Impact force sufficient to cause grievous injuries to rider of Activa.**HELMET EXAMINATION****VICTIM HELMET VEGA****Outer Shell:** Multiple impact marks on right temporal and parietal region. Deep gouge 4cm long on right side**Inner Liner:** Compressed on right side indicating absorption of impact energy

Chin Strap: Intact and fastened at time of impact (tear pattern confirms)

Blood: Blood stains inside helmet, Group B+ confirmed

Hair: Scalp hair fragments with root, suitable for DNA

Cracks: Two linear cracks on outer shell right side, 8cm and 5cm, indicating impact absorption

Effectiveness: Helmet prevented direct skull fracture. Damage pattern consistent with head impact on road surface

ACCUSED HELMET STEELBIRD

Outer Shell: Minor scratches on left side

Inner Liner: Normal, no compression

Chin Strap: Found torn - examination reveals it was already torn/damaged before accident, not due to accident forces

Conclusion: Helmet not properly secured at time of accident. Scratches consistent with minor fall.

BLOOD ANALYSIS

VICTIM BLOOD BLD 0452 B

Blood Group: B+

Alcohol: Not detected (0 mg/dL)

Drugs Toxicology: Negative for all screened substances

Dna Profile: Extracted and preserved for identification

ACCUSED BLOOD BLD 0452 A

Blood Group: O+

Alcohol: Not detected (0 mg/dL)

Drugs Toxicology: Negative for all screened substances

Dna Profile: Extracted and preserved for identification

ACCIDENT RECONSTRUCTION

Method: Forensic analysis of physical evidence, CCTV footage correlation, mathematical modeling

Pre Impact Scenario: Pulsar traveling SW to NE on Shankar Road towards junction. Activa traveling from Rajinder Nagar side and executing right turn towards Old Rajinder Nagar.

Impact Configuration: Pulsar front collides with right center of Activa at intersection. Impact angle 75 degrees.

Post Impact Dynamics: Activa rider thrown to right side, landing 4.2m from impact point. Pulsar falls to right side, slides 3.1m. Pulsar rider falls to left, minimal travel distance.

Speed At Impact: Pulsar: 52-58 kmph (calculated from skid mark length 4.2m, friction coefficient 0.7 for dry bitumen, and residual impact velocity from damage analysis). Activa: 8-12 kmph (typical turning speed).

Avoidability: At speed of 40 kmph (legal limit), Pulsar could have stopped in 15.5m total (thinking + braking distance). From skid mark evidence, Pulsar was at least 8m from impact point when emergency braking initiated. At legal speed, collision was avoidable. At actual speed (55 kmph), stopping distance required was 25m, making collision unavoidable from point where braking started.

Violation: Speed exceeding legal limit was the primary causative factor. Signal violation/secondary as signal was non-functional but traffic police manual direction was ignored.

CCTV ANALYSIS

SOURCES ANALYZED

Item 1:

Camera: St. Stephen's Hospital Gate 2 (CAM-SSH-G2-03)

Timestamp Synced: True

KEY OBSERVATIONS

Item 1:

18:20:03 - White Activa enters frame from bottom (Rajinder Nagar side), speed low, indicator blinking for right turn

Item 2:

18:20:05 - Red Pulsar enters frame from left (Shankar Road, Karol Bagh side), speed visibly high

Item 3:

18:20:06 - Traffic constable visible at junction, left hand raised signaling STOP for Shankar Road traffic

Item 4:

18:20:07 - Pulsar does not stop, continues at same speed into junction

Item 5:

18:20:07.5 - Pulsar front collides with right side of Activa

Item 6:

18:20:08 - Activa rider airborne, falls near road edge

Item 7:

18:20:09 - Pulsar rider falls, gets up at 18:20:12

Item 8:

18:20:14 - Constable runs towards Pulsar rider

Item 9:

18:20:18 - Constable grabs Pulsar rider by arm

Item 10:

Time stamp verified: Camera clock accurate within 2 seconds of IST

Item 2:

Camera: Canara Bank ATM (CAM-CB-ATM-01)

Timestamp Synced: True

KEY OBSERVATIONS

Item 1:

18:19:45 - Red Pulsar passes ATM at high speed, rider wearing red t-shirt, delivery bag visible

Item 2:

18:19:52 - Pulsar continues towards junction (towards St. Stephen's Hospital), brake light not visible at any point

Item 3:

18:20:15 - Sound of crash audible in ATM area (microphone recording), people near ATM look in direction of junction

Item 4:

18:20:30 - Several people seen running towards junction

Conclusion: CCTV footage corroborates witness statements. Confirms overspeeding, disregard of traffic police signal, and sequence of collision. Time stamps consistent with timeline.

FINAL OPINION

Cause Of Accident: The accident occurred primarily due to overspeeding by the rider of Bajaj Pulsar DL2SBP1234 (speed 52-58 kmph against limit of 40 kmph) and failure to obey manual traffic police signal to stop. The speed made it impossible for the rider to stop or avoid collision when the Activa legally executed its turn.

CONTRIBUTING FACTORS

Item 1:

Non-functional traffic signal (administrative failure)

Item 2:

Accused's hurry to deliver food order creating time pressure

Item 3:

Non-functional street lights reducing optimal visibility at dusk

EXONERATING FACTORS VICTIM**Item 1:**

Victim riding at appropriate speed

Item 2:

Victim wearing properly secured ISI-mark helmet

Item 3:

Victim taking legal turn with indicator

Item 4:

No alcohol/drugs detected

Item 5:

No contributory negligence found

INCULPATING FACTORS ACCUSED**Item 1:**

Gross overspeeding (52-58 kmph in 40 kmph zone)

Item 2:

Disregard of traffic constable's manual stop signal

Item 3:

Damaged helmet strap (prior to accident) indicating casual approach to safety

Item 4:

Self-admitted hurry to meet delivery deadline

Signature: Dr. Priya Nambiar, Senior Scientific Officer (Physics), FSL Rohini, Delhi

Seal: Forensic Science Laboratory, Government of NCT of Delhi